

Ibon Gracia

PhD Student in Aerospace Engineering Sciences (Autonomous Systems)

✉ ibon.gracia@colorado.edu

📍 Boulder, Colorado, USA

🔗 [Website](#)

🏠 [Google Scholar](#)

🌐 [LinkedIn](#)

🧠 Research Interests

Safe and assured autonomy
Stochastic systems

Robotics and motion planning
Data-driven control and verification

Formal methods for control

🎓 Education

2023 – Present
Boulder, CO, USA

PhD in Aerospace Engineering Sciences
University of Colorado Boulder (CU Boulder)
Focus: Autonomous Systems
Advisor: Morteza Lahijanian
Comprehensive exam date: 11/2026
Expected graduation date: 05/2027

09/2020 – 10/2022
Delft, The Netherlands

MSc in Systems and Control
Delft University of Technology (TU Delft)
Thesis advisors: Dimitris Boskos, Luca Laurenti

09/2016 – 09/2019
Madrid, Spain

BSc in Aerospace Engineering
Technical University of Madrid (UPM)

09/2012 – 09/2016
Madrid, Spain

BSc in Mechanical Engineering
Technical University of Madrid (UPM)

📄 Publications

Journal Publications

- **Gracia, I.**, & Lahijanian, M. (2025). Beyond Interval MDPs: Tight and Efficient Abstractions of Stochastic Systems. *arXiv preprint arXiv:2507.02213*. Provisionally accepted in *Automatica*
- **Gracia, I.**, Boskos, D., Lahijanian, M., Laurenti, L., & Mazo Jr, M. (2025). Efficient strategy synthesis for switched stochastic systems with distributional uncertainty. *Nonlinear Analysis: Hybrid Systems*, 55, 101554.

Conference Publications

- **Gracia, I.**, & Lahijanian, M. (2026). On the Optimality of Uncertain MDP Abstractions. *arXiv preprint arXiv:2604.13377*. To appear in IEEE Conference for Decision and Control (CDC 2026)
- Kong, C. W., Escobar, S., **Gracia, I.**, McMahon, J., & Lahijanian, M. (2026). Training with hard constraints: Learning neural certificates and controllers for sdes. *arXiv preprint arXiv:2602.23526*.
- **Gracia, I.**, & Lahijanian, M. (2025). Data-Driven Control via Conditional Mean Embeddings: Formal Guarantees via Uncertain MDP Abstraction. *arXiv preprint arXiv:2512.13940*. To appear in ACM International Conference on Hybrid Systems: Computation and Control
- **Gracia, I.**, Boskos, D., Laurenti, L., & Mazo Jr, M. (2023, May). Distributionally robust strategy synthesis for switched stochastic systems. In *Proceedings of the 26th ACM International Conference on Hybrid Systems: Computation and Control* (pp. 1-10). **Best paper award**
- **Gracia, I.**, Boskos, D., Laurenti, L., & Lahijanian, M. (2024, June). Data-driven strategy synthesis for stochastic systems with unknown nonlinear disturbances. In *6th Annual Learning for Dynamics & Control Conference* (pp. 1633-1645). PMLR.

- **Gracia, I.**, Laurenti, L., Mazo Jr., M., Abate, A. and Lahijanian, M. (2025). Temporal Logic Control for Nonlinear Stochastic Systems Under Unknown Disturbances. Proceedings of the 7th Annual Learning for Dynamics & Control Conference. Proceedings of Machine Learning Research

Manuscripts Under Review

- **Gracia, I.**, Ho, Q. H., Laurenti, L., & Lahijanian, M. (2026). Provably Safe Motion Planning Under Unknown Disturbances. *arXiv preprint arXiv:2605.26625*. Submitted for publication in IEEE Transactions in Robotics

Awards

2023 **Best Paper Award**
ACM International Conference on Hybrid Systems: Computation and Control

Teaching Experience

05/2026 – 07/2026 University of Colorado Boulder	ASEN 3728 — Aircraft Dynamics, University of Colorado Boulder Summer 2026 Instructor of record
04/2026 – 04/2026 Boulder, CO, USA	ASEN 6519 — Safe Autonomy Amid Uncertainty University of Colorado Boulder Guest lecturer
01/2026 – 05/2026 Boulder, CO, USA	ASEN 3728: Aircraft Dynamics University of Colorado Boulder Teaching Assistant
08/2026 – Present Boulder, CO, USA	ASEN 5044: Statistical Estimation for Dynamical Systems University of Colorado Boulder Teaching Assistant
08/2025 – 12/2025 Boulder, CO, USA	ASEN 3711: Aerodynamics University of Colorado Boulder Teaching Assistant
07/2021 – 09/2021 Delft, The Netherlands	SC42155: Modelling of Dynamical Systems Delft University of Technology (TU Delft) Student Assistant Helped the professors to extend the reader of a course in system modeling with additional theory and exercises.

Industry Experience

03/2019 – 08/2020 Madrid, Spain	Capgemini DEMS Electronic Flight Control Systems Engineer R&D project on the automation of an air-to-air refueling system. Duties involving flight control systems specification, modeling, verification and validation, hardware-in-the-loop testing, as well as support to departments such as Software, Flight Dynamics and Flight Test.
------------------------------------	--

Conference Presentations

- "Data-Driven Control via Conditional Mean Embeddings: Formal Guarantees via Uncertain MDP Abstraction," *ACM HSCC* 2026
- "Training with Hard Constraints: Learning Neural Certificates and Controllers for SDEs," *NeuS*, 2026
- "Temporal Logic Control for Nonlinear Stochastic Systems Under Unknown Disturbances," *L4DC*, 2025. **Poster**
- ""Data-driven strategy synthesis for stochastic systems with unknown nonlinear disturbances", *L4DC* 2024. **Poster**
- "Distributionally robust strategy synthesis for switched stochastic systems", *ACM HSCC* 2023

Professional Service

Reviewer for Automatica, Hybrid Systems: Computation and Control (HSCC), Conference on Decision and Control (CDC), Learning for Dynamics and Control (L4DC), ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), American Control Conference (ACC), European Control Conference (ECC), IEEE International Conference on Robotics and Automation (ICRA)

Programming Skills

- | | | |
|-------------------|---------|---------|
| • Matlab/Simulink | • C++ | • LaTeX |
| • Python | • Julia | |

Languages

English

Spanish